PG No. 15 $D_{4\mathrm{h}}$ $4/mmm$ [tetragonal] (lg basis)

bra: $= \langle s|$
ket: $= |s\rangle$

Table 1: (s,s) block.

No.	multipole	matrix
1	symmetry	1
	$\mathbb{Q}_0^{(a)}(A_{1g})$	$\begin{bmatrix} 1 \end{bmatrix}$

bra: $= \langle s|$
ket: $= |p_x\rangle, |p_y\rangle, |p_z\rangle$

Table 2: (s,p) block.

No.	multipole	matrix
2	symmetry	z
	$\mathbb{Q}_1^{(a)}(A_{2u})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$
3	symmetry	x
	$\mathbb{Q}_{1,1}^{(a)}(E_u)$	$\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$
4	symmetry	y
	$\mathbb{Q}_{1,2}^{(a)}(E_u)$	$\begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$
5	symmetry	z
	$\mathbb{T}_1^{(a)}(A_{2u})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} \end{bmatrix}$
6	symmetry	x
	$\mathbb{T}_{1,1}^{(a)}(E_u)$	$\begin{bmatrix} \frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$
7	symmetry	y
	$\mathbb{T}_{1,2}^{(a)}(E_u)$	$\begin{bmatrix} 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$

bra: $= \langle s|$
ket: $= |d_v\rangle, |d_{xy}\rangle, |d_{xz}\rangle, |d_{yz}\rangle, |d_u\rangle$

Table 3: (s,d) block.

No.	multipole	matrix
8	symmetry	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
	$\mathbb{Q}_2^{(a)}(A_{1g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$
9	symmetry	$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$
	$\mathbb{Q}_2^{(a)}(B_{1g})$	$\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$
10	symmetry	$\sqrt{3}xy$
	$\mathbb{Q}_2^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 \end{bmatrix}$
11	symmetry	$\sqrt{3}yz$
	$\mathbb{Q}_{2,1}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$
12	symmetry	$\sqrt{3}xz$
	$\mathbb{Q}_{2,2}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$
13	symmetry	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
	$\mathbb{T}_2^{(a)}(A_{1g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \end{bmatrix}$
14	symmetry	$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$
	$\mathbb{T}_2^{(a)}(B_{1g})$	$\begin{bmatrix} \frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$
15	symmetry	$\sqrt{3}xy$
	$\mathbb{T}_2^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & \frac{\sqrt{2}i}{2} & 0 & 0 & 0 \end{bmatrix}$
16	symmetry	$\sqrt{3}yz$
	$\mathbb{T}_{2,1}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$
17	symmetry	$\sqrt{3}xz$
	$\mathbb{T}_{2,2}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$

bra: $= \langle s|$

ket: $= |f_c\rangle, |f_s\rangle, |f_z^\beta\rangle, |f_{xyz}\rangle, |f_x\rangle, |f_y\rangle, |f_z^\alpha\rangle$

Table 4: (s,f) block.

No.	multipole	matrix
18	symmetry	$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$

continued ...

Table 4

No.	multipole	matrix
	$\mathbb{Q}_3^{(a)}(A_{2u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$
19	symmetry	$\sqrt{15}xyz$
	$\mathbb{Q}_3^{(a)}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 \end{bmatrix}$
20	symmetry	$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$
	$\mathbb{Q}_3^{(a)}(B_{2u})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$
21	symmetry	$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$
	$\mathbb{Q}_{3,1}^{(a)}(E_u,1)$	$\begin{bmatrix} \frac{\sqrt{5}}{4} & 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & 0 \end{bmatrix}$
22	symmetry	$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$
	$\mathbb{Q}_{3,2}^{(a)}(E_u,1)$	$\begin{bmatrix} 0 & -\frac{\sqrt{5}}{4} & 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 \end{bmatrix}$
23	symmetry	$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$
	$\mathbb{Q}_{3,1}^{(a)}(E_u,2)$	$\begin{bmatrix} -\frac{\sqrt{3}}{4} & 0 & 0 & 0 & -\frac{\sqrt{5}}{4} & 0 & 0 \end{bmatrix}$
24	symmetry	$\frac{\sqrt{15}x^2y}{2} - \frac{\sqrt{15}yz^2}{2}$
	$\mathbb{Q}_{3,2}^{(a)}(E_u,2)$	$\begin{bmatrix} 0 & \frac{\sqrt{3}}{4} & 0 & 0 & 0 & -\frac{\sqrt{5}}{4} & 0 \end{bmatrix}$
25	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{T}_3^{(a)}(A_{2u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \end{bmatrix}$
26	symmetry	$\sqrt{15}xyz$
	$\mathbb{T}_3^{(a)}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{2} & 0 & 0 & 0 \end{bmatrix}$
27	symmetry	$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$
	$\mathbb{T}_3^{(a)}(B_{2u})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$
28	symmetry	$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$
	$\mathbb{T}_{3,1}^{(a)}(E_u,1)$	$\begin{bmatrix} \frac{\sqrt{5}i}{4} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{4} & 0 & 0 \end{bmatrix}$
29	symmetry	$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$
	$\mathbb{T}_{3,2}^{(a)}(E_u,1)$	$\begin{bmatrix} 0 & -\frac{\sqrt{5}i}{4} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{4} & 0 \end{bmatrix}$
30	symmetry	$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$
	$\mathbb{T}_{3,1}^{(a)}(E_u,2)$	$\begin{bmatrix} -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{4} & 0 & 0 \end{bmatrix}$
31	symmetry	$\frac{\sqrt{15}x^2y}{2} - \frac{\sqrt{15}yz^2}{2}$
	$\mathbb{T}_{3,2}^{(a)}(E_u,2)$	$\begin{bmatrix} 0 & \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{4} & 0 \end{bmatrix}$

bra: $= \langle p_x| , \langle p_y| , \langle p_z|$
ket: $= |p_x\rangle , |p_y\rangle , |p_z\rangle$

Table 5: (p,p) block.

No.	multipole	matrix
32	symmetry	1
	$\mathbb{Q}_0^{(a)}(A_{1g})$	$\begin{bmatrix} \frac{\sqrt{3}}{3} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$
33	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{Q}_2^{(a)}(A_{1g})$	$\begin{bmatrix} -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{3} \end{bmatrix}$
34	symmetry	$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$
	$\mathbb{Q}_2^{(a)}(B_{1g})$	$\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}$
35	symmetry	$\sqrt{3}xy$
	$\mathbb{Q}_2^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
36	symmetry	$\sqrt{3}yz$
	$\mathbb{Q}_{2,1}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$
37	symmetry	$\sqrt{3}xz$
	$\mathbb{Q}_{2,2}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$
38	symmetry	z

continued ...

Table 5

No.	multipole	matrix
	$\mathbb{M}_1^{(a)}(A_{2g})$	$\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} & 0 \\ \frac{\sqrt{2}i}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
39	symmetry	x
	$\mathbb{M}_{1,1}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$
40	symmetry	$-y$
	$\mathbb{M}_{1,2}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 \\ \frac{\sqrt{2}i}{2} & 0 & 0 \end{bmatrix}$

bra: $= \langle p_x| , \langle p_y| , \langle p_z|$

ket: $= |d_v\rangle , |d_{xy}\rangle , |d_{xz}\rangle , |d_{yz}\rangle , |d_u\rangle$

Table 6: (p,d) block.

No.	multipole	matrix
41	symmetry	z
	$\mathbb{Q}_1^{(a)}(A_{2u})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} \end{bmatrix}$
42	symmetry	x
	$\mathbb{Q}_{1,1}^{(a)}(E_u)$	$\begin{bmatrix} \frac{\sqrt{15}}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 \end{bmatrix}$
43	symmetry	y

continued ...

Table 6

No.	multipole	matrix
	$\mathbb{Q}^{(a)}_{1,2}(E_u)$	$\begin{bmatrix} 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 \end{bmatrix}$
44	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{Q}^{(a)}_{3}(A_{2u})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{10} \end{bmatrix}$
45	symmetry	$\sqrt{15}xyz$
	$\mathbb{Q}^{(a)}_{3}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \end{bmatrix}$
46	symmetry	$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$
	$\mathbb{Q}^{(a)}_{3}(B_{2u})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \end{bmatrix}$
47	symmetry	$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$
	$\mathbb{Q}^{(a)}_{3,1}(E_u,1)$	$\begin{bmatrix} \frac{3\sqrt{10}}{20} & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 \end{bmatrix}$
48	symmetry	$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$
	$\mathbb{Q}^{(a)}_{3,2}(E_u,1)$	$\begin{bmatrix} 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ -\frac{3\sqrt{10}}{20} & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 \end{bmatrix}$
49	symmetry	$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$
	$\mathbb{Q}^{(a)}_{3,1}(E_u,2)$	$\begin{bmatrix} -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \end{bmatrix}$
50	symmetry	$\frac{\sqrt{15}x^2y}{2} - \frac{\sqrt{15}yz^2}{2}$

continued ...

Table 6

No.	multipole	matrix
	$\mathbb{Q}_{3,2}^{(a)}(E_u,2)$	$\begin{bmatrix} 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \end{bmatrix}$
51	symmetry	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
	$\mathbb{G}_2^{(a)}(A_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
52	symmetry	$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$
	$\mathbb{G}_2^{(a)}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{3} & 0 & 0 & 0 \end{bmatrix}$
53	symmetry	$\sqrt{3}xy$
	$\mathbb{G}_2^{(a)}(B_{2u})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}}{3} & 0 & 0 & 0 & 0 \end{bmatrix}$
54	symmetry	$\sqrt{3}yz$
	$\mathbb{G}_{2,1}^{(a)}(E_u)$	$\begin{bmatrix} -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \end{bmatrix}$
55	symmetry	$-\sqrt{3}xz$
	$\mathbb{G}_{2,2}^{(a)}(E_u)$	$\begin{bmatrix} 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & 0 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \end{bmatrix}$
56	symmetry	z
	$\mathbb{T}_1^{(a)}(A_{2u})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} \end{bmatrix}$
57	symmetry	x

continued ...

Table 6

No.	multipole	matrix
	$\mathbb{T}^{(a)}_{1,1}(E_u)$	$\begin{bmatrix} \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 \end{bmatrix}$
58	symmetry	y
	$\mathbb{T}^{(a)}_{1,2}(E_u)$	$\begin{bmatrix} 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 \end{bmatrix}$
59	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{T}^{(a)}_{3}(A_{2u})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{10} \end{bmatrix}$
60	symmetry	$\sqrt{15}xyz$
	$\mathbb{T}^{(a)}_{3}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \end{bmatrix}$
61	symmetry	$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$
	$\mathbb{T}^{(a)}_{3}(B_{2u})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \end{bmatrix}$
62	symmetry	$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$
	$\mathbb{T}^{(a)}_{3,1}(E_u,1)$	$\begin{bmatrix} \frac{3\sqrt{10}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \end{bmatrix}$
63	symmetry	$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$
	$\mathbb{T}^{(a)}_{3,2}(E_u,1)$	$\begin{bmatrix} 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{3\sqrt{10}i}{20} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \end{bmatrix}$
64	symmetry	$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$

continued ...

Table 6

No.	multipole	matrix
	$\mathbb{T}_{3,1}^{(a)}(E_u,2)$	$\begin{bmatrix} -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \end{bmatrix}$
65	symmetry	$\frac{\sqrt{15}x^2y}{2} - \frac{\sqrt{15}yz^2}{2}$
	$\mathbb{T}_{3,2}^{(a)}(E_u,2)$	$\begin{bmatrix} 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$
66	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{M}_2^{(a)}(A_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
67	symmetry	$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$
	$\mathbb{M}_2^{(a)}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{3} & 0 & 0 & 0 \end{bmatrix}$
68	symmetry	$\sqrt{3}xy$
	$\mathbb{M}_2^{(a)}(B_{2u})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ -\frac{\sqrt{3}i}{3} & 0 & 0 & 0 & 0 \end{bmatrix}$
69	symmetry	$\sqrt{3}yz$
	$\mathbb{M}_{2,1}^{(a)}(E_u)$	$\begin{bmatrix} \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & \frac{i}{2} \\ 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \end{bmatrix}$
70	symmetry	$-\sqrt{3}xz$
	$\mathbb{M}_{2,2}^{(a)}(E_u)$	$\begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$

bra: $= \langle p_x| , \langle p_y| , \langle p_z|$

ket: $= |f_c\rangle, |f_s\rangle, |f_z^\beta\rangle, |f_{xyz}\rangle, |f_x\rangle, |f_y\rangle, |f_z^\alpha\rangle$

Table 7: (p,f) block.

No.	multipole	matrix
71	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{Q}_2^{(a)}(A_{1g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} \end{bmatrix}$
72	symmetry	$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$
	$\mathbb{Q}_2^{(a)}(B_{1g})$	$\begin{bmatrix} \frac{\sqrt{35}}{14} & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 \\ 0 & \frac{\sqrt{35}}{14} & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 \end{bmatrix}$
73	symmetry	$\sqrt{3}xy$
	$\mathbb{Q}_2^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & \frac{\sqrt{35}}{14} & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ -\frac{\sqrt{35}}{14} & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 \end{bmatrix}$
74	symmetry	$\sqrt{3}yz$
	$\mathbb{Q}_{2,1}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{42} & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{21} & 0 \end{bmatrix}$
75	symmetry	$\sqrt{3}xz$
	$\mathbb{Q}_{2,2}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{21} & 0 & 0 \end{bmatrix}$
76	symmetry	$\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$
	$\mathbb{Q}_4^{(a)}(A_{1g}, 1)$	$\begin{bmatrix} \frac{\sqrt{15}}{12} & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$
77	symmetry	$-\frac{\sqrt{15}x^4}{12} + \sqrt{15}x^2y^2 - \frac{\sqrt{15}x^2z^2}{2} - \frac{\sqrt{15}y^4}{12} - \frac{\sqrt{15}y^2z^2}{2} + \frac{\sqrt{15}z^4}{6}$

continued ...

Table 7

No.	multipole	matrix
	$\mathbb{Q}_4^{(a)}(A_{1g},2)$	$\begin{bmatrix} -\frac{\sqrt{21}}{12} & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 \\ 0 & \frac{\sqrt{21}}{12} & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{42} \end{bmatrix}$
78	symmetry	$\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$
	$\mathbb{Q}_4^{(a)}(A_{2g})$	$\begin{bmatrix} 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
79	symmetry	$\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$
	$\mathbb{Q}_4^{(a)}(B_{1g})$	$\begin{bmatrix} \frac{\sqrt{7}}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 & 0 \\ 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{105}}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$
80	symmetry	$-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$
	$\mathbb{Q}_4^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{105}}{28} & 0 \\ \frac{\sqrt{7}}{28} & 0 & 0 & 0 & \frac{\sqrt{105}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \end{bmatrix}$
81	symmetry	$\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$
	$\mathbb{Q}_{4,1}^{(a)}(E_g,1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} \\ 0 & -\frac{1}{8} & 0 & 0 & 0 & -\frac{\sqrt{15}}{8} & 0 \end{bmatrix}$
82	symmetry	$\frac{\sqrt{35}x^3z}{2} - \frac{\sqrt{35}xz^3}{2}$
	$\mathbb{Q}_{4,2}^{(a)}(E_g,1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{8} & 0 & 0 & 0 & -\frac{\sqrt{15}}{8} & 0 & 0 \end{bmatrix}$
83	symmetry	$3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$
	$\mathbb{Q}_{4,1}^{(a)}(E_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} \\ 0 & \frac{\sqrt{7}}{8} & 0 & 0 & 0 & -\frac{\sqrt{105}}{56} & 0 \end{bmatrix}$
84	symmetry	$-\frac{\sqrt{5}x^3z}{2} + 3\sqrt{5}xy^2z - \frac{\sqrt{5}xz^3}{2}$

continued ...

Table 7

No.	multipole	matrix
	$\mathbb{Q}^{(a)}_{4,2}(E_g,2)$	$\begin{bmatrix} 0 & 0 & -\frac{3\sqrt{42}}{56} & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{8} & 0 & 0 & 0 & -\frac{\sqrt{105}}{56} & 0 & 0 \end{bmatrix}$
85	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{G}^{(a)}_{3}(A_{2g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
86	symmetry	$\sqrt{15}xyz$
	$\mathbb{G}^{(a)}_{3}(B_{1g})$	$\begin{bmatrix} -\frac{1}{4} & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 \\ 0 & -\frac{1}{4} & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \end{bmatrix}$
87	symmetry	$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$
	$\mathbb{G}^{(a)}_{3}(B_{2g})$	$\begin{bmatrix} 0 & \frac{1}{4} & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 \\ -\frac{1}{4} & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 \end{bmatrix}$
88	symmetry	$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$
	$\mathbb{G}^{(a)}_{3,1}(E_g,1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} \\ 0 & -\frac{\sqrt{15}}{8} & 0 & 0 & 0 & \frac{1}{8} & 0 \end{bmatrix}$
89	symmetry	$\frac{3x^2y}{2} - y^3 + \frac{3yz^2}{2}$
	$\mathbb{G}^{(a)}_{3,2}(E_g,1)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}}{8} & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{8} & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 \end{bmatrix}$
90	symmetry	$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$
	$\mathbb{G}^{(a)}_{3,1}(E_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} \\ 0 & \frac{3}{8} & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 \end{bmatrix}$
91	symmetry	$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$

continued ...

Table 7

No.	multipole	matrix
	$\mathbb{G}_{3,2}^{(a)}(E_g,2)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{3}{8} & 0 & 0 & 0 & \frac{\sqrt{15}}{24} & 0 & 0 \end{bmatrix}$
92	symmetry	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
	$\mathbb{T}_2^{(a)}(A_{1g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} \end{bmatrix}$
93	symmetry	$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$
	$\mathbb{T}_2^{(a)}(B_{1g})$	$\begin{bmatrix} \frac{\sqrt{35}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{14} & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 \end{bmatrix}$
94	symmetry	$\sqrt{3}xy$
	$\mathbb{T}_2^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & \frac{\sqrt{35}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 \\ -\frac{\sqrt{35}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 \end{bmatrix}$
95	symmetry	$\sqrt{3}yz$
	$\mathbb{T}_{2,1}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{21} & 0 \end{bmatrix}$
96	symmetry	$\sqrt{3}xz$
	$\mathbb{T}_{2,2}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{21} & 0 & 0 \end{bmatrix}$
97	symmetry	$\frac{\sqrt{21}x^4}{6}-\frac{\sqrt{21}x^2y^2}{2}-\frac{\sqrt{21}x^2z^2}{2}+\frac{\sqrt{21}y^4}{6}-\frac{\sqrt{21}y^2z^2}{2}+\frac{\sqrt{21}z^4}{6}$
	$\mathbb{T}_4^{(a)}(A_{1g},1)$	$\begin{bmatrix} \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & -\frac{i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} \end{bmatrix}$
98	symmetry	$-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$

continued ...

Table 7

No.	multipole	matrix
	$\mathbb{T}_4^{(a)}(A_{1g},2)$	$\begin{bmatrix} -\frac{\sqrt{21}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{12} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{42} \end{bmatrix}$
99	symmetry	$\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$
	$\mathbb{T}_4^{(a)}(A_{2g})$	$\begin{bmatrix} 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
100	symmetry	$\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$
	$\mathbb{T}_4^{(a)}(B_{1g})$	$\begin{bmatrix} \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 & 0 \\ 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$
101	symmetry	$-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$
	$\mathbb{T}_4^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 \\ \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 \end{bmatrix}$
102	symmetry	$\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$
	$\mathbb{T}_{4,1}^{(a)}(E_g,1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} \\ 0 & -\frac{i}{8} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{8} & 0 \end{bmatrix}$
103	symmetry	$\frac{\sqrt{35}x^3z}{2} - \frac{\sqrt{35}xz^3}{2}$
	$\mathbb{T}_{4,2}^{(a)}(E_g,1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{8} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{8} & 0 & 0 \end{bmatrix}$
104	symmetry	$3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$
	$\mathbb{T}_{4,1}^{(a)}(E_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{42}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} \\ 0 & \frac{\sqrt{7}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{56} & 0 \end{bmatrix}$
105	symmetry	$-\frac{\sqrt{5}x^3z}{2} + 3\sqrt{5}xy^2z - \frac{\sqrt{5}xz^3}{2}$

continued ...

Table 7

No.	multipole	matrix
	$\mathbb{T}^{(a)}_{4,2}(E_g,2)$	$\begin{bmatrix} 0 & 0 & -\frac{3\sqrt{42}i}{56} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{105}i}{56} & 0 & 0 \end{bmatrix}$
106	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{M}^{(a)}_{3}(A_{2g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
107	symmetry	$\sqrt{15}xyz$
	$\mathbb{M}^{(a)}_{3}(B_{1g})$	$\begin{bmatrix} \frac{i}{4} & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 \\ 0 & \frac{i}{4} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \end{bmatrix}$
108	symmetry	$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$
	$\mathbb{M}^{(a)}_{3}(B_{2g})$	$\begin{bmatrix} 0 & -\frac{i}{4} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 \\ \frac{i}{4} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \end{bmatrix}$
109	symmetry	$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$
	$\mathbb{M}^{(a)}_{3,1}(E_g,1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} \\ 0 & \frac{\sqrt{15}i}{8} & 0 & 0 & 0 & -\frac{i}{8} & 0 \end{bmatrix}$
110	symmetry	$\frac{3x^2y}{2} - y^3 + \frac{3yz^2}{2}$
	$\mathbb{M}^{(a)}_{3,2}(E_g,1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{10}i}{8} & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{8} & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 \end{bmatrix}$
111	symmetry	$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$
	$\mathbb{M}^{(a)}_{3,1}(E_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & \frac{\sqrt{10}i}{8} \\ 0 & -\frac{3i}{8} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 \end{bmatrix}$
112	symmetry	$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$

continued ...

Table 7

No.	multipole	matrix
	$\mathbb{M}^{(a)}_{3,2}(E_g,2)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & \frac{\sqrt{10}i}{8} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ \frac{3i}{8} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{24} & 0 & 0 \end{bmatrix}$

bra: $= \langle d_v|, \langle d_{xy}|, \langle d_{xz}|, \langle d_{yz}|, \langle d_u|$
ket: $= |d_v\rangle, |d_{xy}\rangle, |d_{xz}\rangle, |d_{yz}\rangle, |d_u\rangle$

Table 8: (d,d) block.

No.	multipole	matrix
113	symmetry	1
	$\mathbb{Q}^{(a)}_0(A_{1g})$	$\begin{bmatrix} \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{5} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} \end{bmatrix}$
114	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{Q}^{(a)}_2(A_{1g})$	$\begin{bmatrix} -\frac{\sqrt{14}}{7} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{7} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{7} \end{bmatrix}$
115	symmetry	$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$
	$\mathbb{Q}^{(a)}_2(B_{1g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{7} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ -\frac{\sqrt{14}}{7} & 0 & 0 & 0 & 0 \end{bmatrix}$
116	symmetry	$\sqrt{3}xy$

continued ...

Table 8

No.	multipole	matrix
	$\mathbb{Q}_2^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{7} \\ 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{7} & 0 & 0 & 0 \end{bmatrix}$
117	symmetry	$\sqrt{3}yz$
	$\mathbb{Q}_{2,1}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \end{bmatrix}$
118	symmetry	$\sqrt{3}xz$
	$\mathbb{Q}_{2,2}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 \\ \frac{\sqrt{42}}{14} & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \end{bmatrix}$
119	symmetry	$\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$
	$\mathbb{Q}_4^{(a)}(A_{1g},1)$	$\begin{bmatrix} \frac{\sqrt{30}}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{15} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{15} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{10} \end{bmatrix}$
120	symmetry	$-\frac{\sqrt{15}x^4}{12} + \sqrt{15}x^2y^2 - \frac{\sqrt{15}x^2z^2}{2} - \frac{\sqrt{15}y^4}{12} - \frac{\sqrt{15}y^2z^2}{2} + \frac{\sqrt{15}z^4}{6}$

continued ...

Table 8

No.	multipole	matrix
	$\mathbb{Q}_4^{(a)}(A_{1g},2)$	$\begin{bmatrix} -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{42}}{21} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} \end{bmatrix}$
121	symmetry	$\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$
	$\mathbb{Q}_4^{(a)}(A_{2g})$	$\begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
122	symmetry	$\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$
	$\mathbb{Q}_4^{(a)}(B_{1g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{7} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{7} & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$
123	symmetry	$-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$
	$\mathbb{Q}_4^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{7} & 0 \\ 0 & 0 & \frac{\sqrt{14}}{7} & 0 & 0 \\ 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \end{bmatrix}$
124	symmetry	$\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$

continued ...

Table 8

No.	multipole	matrix
	$\mathbb{Q}_{4,1}^{(a)}(E_g,1)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & -\frac{\sqrt{6}}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{4} & 0 \end{bmatrix}$
125	symmetry	$\frac{\sqrt{35}x^3z}{2} - \frac{\sqrt{35}xz^3}{2}$
	$\mathbb{Q}_{4,2}^{(a)}(E_g,1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & 0 & 0 & -\frac{\sqrt{6}}{4} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{4} & 0 & 0 \end{bmatrix}$
126	symmetry	$3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$
	$\mathbb{Q}_{4,1}^{(a)}(E_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{3\sqrt{14}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{14}}{7} & 0 & 0 \\ 0 & \frac{\sqrt{14}}{7} & 0 & 0 & 0 \\ \frac{3\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 \end{bmatrix}$
127	symmetry	$-\frac{\sqrt{5}x^3z}{2} + 3\sqrt{5}xy^2z - \frac{\sqrt{5}xz^3}{2}$
	$\mathbb{Q}_{4,2}^{(a)}(E_g,2)$	$\begin{bmatrix} 0 & 0 & -\frac{3\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{7} & 0 \\ -\frac{3\sqrt{14}}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & \frac{\sqrt{14}}{7} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 \end{bmatrix}$
128	symmetry	z

continued ...

Table 8

No.	multipole	matrix
	$\mathbb{M}_1^{(a)}(A_{2g})$	$\begin{bmatrix} 0 & -\frac{\sqrt{10}i}{5} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
129	symmetry	x
	$\mathbb{M}_{1,1}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{10} & 0 \end{bmatrix}$
130	symmetry	$-y$
	$\mathbb{M}_{1,2}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{10} \\ 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{10} & 0 & 0 \end{bmatrix}$
131	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{M}_3^{(a)}(A_{2g})$	$\begin{bmatrix} 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{5} & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{5} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
132	symmetry	$\sqrt{15}xyz$

continued ...

Table 8

No.	multipole	matrix
	$\mathbb{M}_3^{(a)}(B_{1g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$
133	symmetry	$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$
	$\mathbb{M}_3^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{2} & 0 & 0 & 0 \end{bmatrix}$
134	symmetry	$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$
	$\mathbb{M}_{3,1}^{(a)}(E_g,1)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{5} & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{5} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 \end{bmatrix}$
135	symmetry	$\frac{3x^2y}{2} - y^3 + \frac{3yz^2}{2}$
	$\mathbb{M}_{3,2}^{(a)}(E_g,1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{5} & 0 \\ -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} \\ 0 & \frac{\sqrt{10}i}{5} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 \end{bmatrix}$
136	symmetry	$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$

continued ...

Table 8

No.	multipole	matrix
	$\mathbb{M}_{3,1}^{(a)}(E_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{4} & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$
137	symmetry	$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$
	$\mathbb{M}_{3,2}^{(a)}(E_g,2)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{4} & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$

bra: $= \langle d_v|, \langle d_{xy}|, \langle d_{xz}|, \langle d_{yz}|, \langle d_u|$
ket: $= |f_c\rangle, |f_s\rangle, |f_z^\beta\rangle, |f_{xyz}\rangle, |f_x\rangle, |f_y\rangle, |f_z^\alpha\rangle$

Table 9: (d,f) block.

No.	multipole	matrix
138	symmetry	z
	$\mathbb{Q}_1^{(a)}(A_{2u})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} \end{bmatrix}$
139	symmetry	x

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{Q}_{1,1}^{(a)}(E_u)$	$\begin{bmatrix} \frac{\sqrt{21}}{14} & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 \\ 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & 0 \end{bmatrix}$
140	symmetry	y
	$\mathbb{Q}_{1,2}^{(a)}(E_u)$	$\begin{bmatrix} 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 \\ -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0 \end{bmatrix}$
141	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{Q}_3^{(a)}(A_{2u})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15} \end{bmatrix}$
142	symmetry	$\sqrt{15}xyz$
	$\mathbb{Q}_3^{(a)}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & \frac{1}{4} & 0 \\ -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
143	symmetry	$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{Q}_3^{(a)}(B_{2u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{12} & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 \\ 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
144	symmetry	$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$
	$\mathbb{Q}_{3,1}^{(a)}(E_u,1)$	$\begin{bmatrix} \frac{1}{8} & 0 & 0 & 0 & -\frac{11\sqrt{15}}{120} & 0 & 0 \\ 0 & \frac{1}{8} & 0 & 0 & 0 & -\frac{\sqrt{15}}{120} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{10}}{40} \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{5\sqrt{3}}{24} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{40} & 0 & 0 \end{bmatrix}$
145	symmetry	$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$
	$\mathbb{Q}_{3,2}^{(a)}(E_u,1)$	$\begin{bmatrix} 0 & \frac{1}{8} & 0 & 0 & 0 & \frac{11\sqrt{15}}{120} & 0 \\ -\frac{1}{8} & 0 & 0 & 0 & -\frac{\sqrt{15}}{120} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{10}}{40} \\ 0 & \frac{5\sqrt{3}}{24} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{40} & 0 \end{bmatrix}$
146	symmetry	$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$
	$\mathbb{Q}_{3,1}^{(a)}(E_u,2)$	$\begin{bmatrix} \frac{\sqrt{15}}{24} & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 \\ 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & -\frac{3}{8} & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{8} & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{8} & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 \end{bmatrix}$
147	symmetry	$\frac{\sqrt{15}x^2y}{2} - \frac{\sqrt{15}yz^2}{2}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{Q}_{3,2}^{(a)}(E_u,2)$	$\begin{bmatrix} 0 & \frac{\sqrt{15}}{24} & 0 & 0 & 0 & \frac{1}{8} & 0 \\ -\frac{\sqrt{15}}{24} & 0 & 0 & 0 & -\frac{3}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} \\ 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 \end{bmatrix}$
148	symmetry	$\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$
	$\mathbb{Q}_5^{(a)}(A_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
149	symmetry	$\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$
	$\mathbb{Q}_5^{(a)}(A_{2u},1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{21} \end{bmatrix}$
150	symmetry	$\frac{3\sqrt{35}x^4z}{8} - \frac{9\sqrt{35}x^2y^2z}{4} + \frac{3\sqrt{35}y^4z}{8}$
	$\mathbb{Q}_5^{(a)}(A_{2u},2)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 \\ \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
151	symmetry	$\frac{\sqrt{105}x^3yz}{2} + \frac{\sqrt{105}xy^3z}{2} - \sqrt{105}xyz^3$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{Q}_5^{(a)}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 \end{bmatrix}$
152	symmetry	$-\frac{\sqrt{105}x^4z}{4} + \frac{\sqrt{105}x^2z^3}{2} + \frac{\sqrt{105}y^4z}{4} - \frac{\sqrt{105}y^2z^3}{2}$
	$\mathbb{Q}_5^{(a)}(B_{2u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 \end{bmatrix}$
153	symmetry	$x^5 - 5x^3y^2 - 5x^3z^2 + \frac{15xy^4}{8} + \frac{15xy^2z^2}{4} + \frac{15xz^4}{8}$
	$\mathbb{Q}_{5,1}^{(a)}(E_u,1)$	$\begin{bmatrix} \frac{11\sqrt{14}}{112} & 0 & 0 & 0 & -\frac{5\sqrt{210}}{336} & 0 & 0 \\ 0 & -\frac{5\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 \\ 0 & 0 & -\frac{5\sqrt{21}}{84} & 0 & 0 & 0 & \frac{\sqrt{35}}{28} \\ 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{48} & 0 & 0 & 0 & \frac{3\sqrt{70}}{112} & 0 & 0 \end{bmatrix}$
154	symmetry	$\frac{15x^4y}{8} - 5x^2y^3 + \frac{15x^2yz^2}{4} + y^5 - 5y^3z^2 + \frac{15yz^4}{8}$
	$\mathbb{Q}_{5,2}^{(a)}(E_u,1)$	$\begin{bmatrix} 0 & \frac{11\sqrt{14}}{112} & 0 & 0 & 0 & \frac{5\sqrt{210}}{336} & 0 \\ \frac{5\sqrt{14}}{56} & 0 & 0 & 0 & \frac{\sqrt{210}}{168} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & \frac{\sqrt{35}}{28} \\ 0 & \frac{\sqrt{42}}{48} & 0 & 0 & 0 & \frac{3\sqrt{70}}{112} & 0 \end{bmatrix}$
155	symmetry	$\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{Q}^{(a)}_{5,1}(E_u,2)$	$\begin{bmatrix} \frac{3\sqrt{10}}{80} & 0 & 0 & 0 & \frac{\sqrt{6}}{16} & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{40} & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 \\ 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 \\ \frac{3\sqrt{30}}{80} & 0 & 0 & 0 & \frac{3\sqrt{2}}{16} & 0 & 0 \end{bmatrix}$
156	symmetry	$\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$
	$\mathbb{Q}^{(a)}_{5,2}(E_u,2)$	$\begin{bmatrix} 0 & \frac{3\sqrt{10}}{80} & 0 & 0 & 0 & -\frac{\sqrt{6}}{16} & 0 \\ \frac{\sqrt{10}}{40} & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{20} & 0 & 0 & 0 & \frac{1}{4} \\ 0 & -\frac{3\sqrt{30}}{80} & 0 & 0 & 0 & \frac{3\sqrt{2}}{16} & 0 \end{bmatrix}$
157	symmetry	$\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$
	$\mathbb{Q}^{(a)}_{5,1}(E_u,3)$	$\begin{bmatrix} -\frac{7\sqrt{30}}{120} & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{40} & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 \end{bmatrix}$
158	symmetry	$-\frac{\sqrt{105}x^4y}{4} + \frac{\sqrt{105}x^2y^3}{2} - \frac{\sqrt{105}y^3z^2}{2} + \frac{\sqrt{105}yz^4}{4}$
	$\mathbb{Q}^{(a)}_{5,2}(E_u,3)$	$\begin{bmatrix} 0 & -\frac{7\sqrt{30}}{120} & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 \\ -\frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & \frac{\sqrt{10}}{40} & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 \end{bmatrix}$
159	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{G}_2^{(a)}(A_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{35}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
160	symmetry	$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$
	$\mathbb{G}_2^{(a)}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 \\ \frac{\sqrt{70}}{28} & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{14} & 0 & 0 & 0 \end{bmatrix}$
161	symmetry	$\sqrt{3}xy$
	$\mathbb{G}_2^{(a)}(B_{2u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{70}}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 \\ 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{35}}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$
162	symmetry	$\sqrt{3}yz$
	$\mathbb{G}_{2,1}^{(a)}(E_u)$	$\begin{bmatrix} -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \end{bmatrix}$
163	symmetry	$-\sqrt{3}xz$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{G}_{2,2}^{(a)}(E_u)$	$\begin{bmatrix} 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 \\ \frac{\sqrt{70}}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \end{bmatrix}$
164	symmetry	$\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$
	$\mathbb{G}_4^{(a)}(A_{1u},1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{4} & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 \\ -\frac{1}{4} & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
165	symmetry	$-\frac{\sqrt{15}x^4}{12} + \sqrt{15}x^2y^2 - \frac{\sqrt{15}x^2z^2}{2} - \frac{\sqrt{15}y^4}{12} - \frac{\sqrt{15}y^2z^2}{2} + \frac{\sqrt{15}z^4}{6}$
	$\mathbb{G}_4^{(a)}(A_{1u},2)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{210}}{35} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{210} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{20} & 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 \\ \frac{\sqrt{35}}{20} & 0 & 0 & 0 & -\frac{5\sqrt{21}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
166	symmetry	$\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$
	$\mathbb{G}_4^{(a)}(A_{2u})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ \frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
167	symmetry	$\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{G}_4^{(a)}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} \\ 0 & -\frac{3\sqrt{105}}{140} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 \\ \frac{3\sqrt{105}}{140} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}}{35} & 0 & 0 & 0 \end{bmatrix}$
168	symmetry	$-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$
	$\mathbb{G}_4^{(a)}(B_{2u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{105}}{140} & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & -\frac{3\sqrt{105}}{140} & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{210}}{35} & 0 & 0 & 0 & 0 \end{bmatrix}$
169	symmetry	$\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$
	$\mathbb{G}_{4,1}^{(a)}(E_u,1)$	$\begin{bmatrix} -\frac{\sqrt{15}}{40} & 0 & 0 & 0 & -\frac{1}{8} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{40} & 0 & 0 & 0 & -\frac{3}{8} & 0 \\ 0 & 0 & \frac{3\sqrt{10}}{40} & 0 & 0 & 0 & \frac{\sqrt{6}}{8} \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ -\frac{3\sqrt{5}}{40} & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 \end{bmatrix}$
170	symmetry	$-\frac{\sqrt{35}x^3z}{2} + \frac{\sqrt{35}xz^3}{2}$
	$\mathbb{G}_{4,2}^{(a)}(E_u,1)$	$\begin{bmatrix} 0 & -\frac{\sqrt{15}}{40} & 0 & 0 & 0 & \frac{1}{8} & 0 \\ \frac{\sqrt{15}}{40} & 0 & 0 & 0 & -\frac{3}{8} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{10}}{40} & 0 & 0 & 0 & \frac{\sqrt{6}}{8} \\ 0 & \frac{3\sqrt{5}}{40} & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 \end{bmatrix}$
171	symmetry	$3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{G}_{4,1}^{(a)}(E_u,2)$	$\begin{bmatrix} -\frac{\sqrt{105}}{280} & 0 & 0 & 0 & -\frac{9\sqrt{7}}{56} & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{280} & 0 & 0 & 0 & \frac{5\sqrt{7}}{56} & 0 \\ 0 & 0 & \frac{\sqrt{70}}{40} & 0 & 0 & 0 & \frac{\sqrt{42}}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{35}}{40} & 0 & 0 & 0 & -\frac{\sqrt{21}}{56} & 0 & 0 \end{bmatrix}$
172	symmetry	$\frac{\sqrt{5}x^3z}{2} - 3\sqrt{5}xy^2z + \frac{\sqrt{5}xz^3}{2}$
	$\mathbb{G}_{4,2}^{(a)}(E_u,2)$	$\begin{bmatrix} 0 & -\frac{\sqrt{105}}{280} & 0 & 0 & 0 & \frac{9\sqrt{7}}{56} & 0 \\ \frac{\sqrt{105}}{280} & 0 & 0 & 0 & \frac{5\sqrt{7}}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{40} & 0 & 0 & 0 & \frac{\sqrt{42}}{56} \\ 0 & -\frac{3\sqrt{35}}{40} & 0 & 0 & 0 & -\frac{\sqrt{21}}{56} & 0 \end{bmatrix}$
173	symmetry	z
	$\mathbb{T}_1^{(a)}(A_{2u})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{35}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} \end{bmatrix}$
174	symmetry	x
	$\mathbb{T}_{1,1}^{(a)}(E_u)$	$\begin{bmatrix} \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{70} \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 \end{bmatrix}$
175	symmetry	y

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{T}^{(a)}_{1,2}(E_u)$	$\begin{bmatrix} 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 \\ -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{70} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 \end{bmatrix}$
176	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{T}^{(a)}_{3}(A_{2u})$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{15} \end{bmatrix}$
177	symmetry	$\sqrt{15}xyz$
	$\mathbb{T}^{(a)}_{3}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & \frac{i}{4} & 0 \\ -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
178	symmetry	$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$
	$\mathbb{T}^{(a)}_{3}(B_{2u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & -\frac{i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
179	symmetry	$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{T}^{(a)}_{3,1}(E_u,1)$	$\begin{bmatrix} \frac{i}{8} & 0 & 0 & 0 & -\frac{11\sqrt{15}i}{120} & 0 & 0 \\ 0 & \frac{i}{8} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{120} & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{40} \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ -\frac{5\sqrt{3}i}{24} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{40} & 0 & 0 \end{bmatrix}$
180	symmetry	$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$
	$\mathbb{T}^{(a)}_{3,2}(E_u,1)$	$\begin{bmatrix} 0 & \frac{i}{8} & 0 & 0 & 0 & \frac{11\sqrt{15}i}{120} & 0 \\ -\frac{i}{8} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{120} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{40} \\ 0 & \frac{5\sqrt{3}i}{24} & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{40} & 0 \end{bmatrix}$
181	symmetry	$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$
	$\mathbb{T}^{(a)}_{3,1}(E_u,2)$	$\begin{bmatrix} \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & -\frac{3i}{8} & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 \end{bmatrix}$
182	symmetry	$\frac{\sqrt{15}x^2y}{2}-\frac{\sqrt{15}yz^2}{2}$
	$\mathbb{T}^{(a)}_{3,2}(E_u,2)$	$\begin{bmatrix} 0 & \frac{\sqrt{15}i}{24} & 0 & 0 & 0 & \frac{i}{8} & 0 \\ -\frac{\sqrt{15}i}{24} & 0 & 0 & 0 & -\frac{3i}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{24} \\ 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 \end{bmatrix}$
183	symmetry	$\frac{3\sqrt{35}x^3yz}{2}-\frac{3\sqrt{35}xy^3z}{2}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{T}_5^{(a)}(A_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
184	symmetry	$\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$
	$\mathbb{T}_5^{(a)}(A_{2u}, 1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{21} \end{bmatrix}$
185	symmetry	$\frac{3\sqrt{35}x^4z}{8} - \frac{9\sqrt{35}x^2y^2z}{4} + \frac{3\sqrt{35}y^4z}{8}$
	$\mathbb{T}_5^{(a)}(A_{2u}, 2)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
186	symmetry	$\frac{\sqrt{105}x^3yz}{2} + \frac{\sqrt{105}xy^3z}{2} - \sqrt{105}xyz^3$
	$\mathbb{T}_5^{(a)}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 \end{bmatrix}$
187	symmetry	$-\frac{\sqrt{105}x^4z}{4} + \frac{\sqrt{105}x^2z^3}{2} + \frac{\sqrt{105}y^4z}{4} - \frac{\sqrt{105}y^2z^3}{2}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{T}^{(a)}_5(B_{2u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 \end{bmatrix}$
188	symmetry	$x^5 - 5x^3y^2 - 5x^3z^2 + \frac{15xy^4}{8} + \frac{15xy^2z^2}{4} + \frac{15xz^4}{8}$
	$\mathbb{T}^{(a)}_{5,1}(E_u,1)$	$\begin{bmatrix} \frac{11\sqrt{14}i}{112} & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{336} & 0 & 0 \\ 0 & -\frac{5\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 \\ 0 & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{48} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{112} & 0 & 0 \end{bmatrix}$
189	symmetry	$\frac{15x^4y}{8} - 5x^2y^3 + \frac{15x^2yz^2}{4} + y^5 - 5y^3z^2 + \frac{15yz^4}{8}$
	$\mathbb{T}^{(a)}_{5,2}(E_u,1)$	$\begin{bmatrix} 0 & \frac{11\sqrt{14}i}{112} & 0 & 0 & 0 & \frac{5\sqrt{210}i}{336} & 0 \\ \frac{5\sqrt{14}i}{56} & 0 & 0 & 0 & \frac{\sqrt{210}i}{168} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} \\ 0 & \frac{\sqrt{42}i}{48} & 0 & 0 & 0 & \frac{3\sqrt{70}i}{112} & 0 \end{bmatrix}$
190	symmetry	$\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$
	$\mathbb{T}^{(a)}_{5,1}(E_u,2)$	$\begin{bmatrix} \frac{3\sqrt{10}i}{80} & 0 & 0 & 0 & \frac{\sqrt{6}i}{16} & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{40} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & \frac{i}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ \frac{3\sqrt{30}i}{80} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{16} & 0 & 0 \end{bmatrix}$
191	symmetry	$\frac{3\sqrt{35}x^4y}{8} - \frac{9\sqrt{35}x^2yz^2}{4} + \frac{3\sqrt{35}yz^4}{8}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{T}^{(a)}_{5,2}(E_u,2)$	$\begin{bmatrix} 0 & \frac{3\sqrt{10}i}{80} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{16} & 0 \\ \frac{\sqrt{10}i}{40} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{20} & 0 & 0 & 0 & \frac{i}{4} \\ 0 & -\frac{3\sqrt{30}i}{80} & 0 & 0 & 0 & \frac{3\sqrt{2}i}{16} & 0 \end{bmatrix}$
192	symmetry	$\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$
	$\mathbb{T}^{(a)}_{5,1}(E_u,3)$	$\begin{bmatrix} -\frac{7\sqrt{30}i}{120} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{40} & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 \end{bmatrix}$
193	symmetry	$-\frac{\sqrt{105}x^4y}{4} + \frac{\sqrt{105}x^2y^3}{2} - \frac{\sqrt{105}y^3z^2}{2} + \frac{\sqrt{105}yz^4}{4}$
	$\mathbb{T}^{(a)}_{5,2}(E_u,3)$	$\begin{bmatrix} 0 & -\frac{7\sqrt{30}i}{120} & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 \\ -\frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & \frac{\sqrt{10}i}{40} & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 \end{bmatrix}$
194	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{M}^{(a)}_{2}(A_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{35}i}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
195	symmetry	$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{M}_2^{(a)}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 \\ -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{14} & 0 & 0 & 0 \end{bmatrix}$
196	symmetry	$\sqrt{3}xy$
	$\mathbb{M}_2^{(a)}(B_{2u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{35}i}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$
197	symmetry	$\sqrt{3}yz$
	$\mathbb{M}_{2,1}^{(a)}(E_u)$	$\begin{bmatrix} \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{7} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 \end{bmatrix}$
198	symmetry	$-\sqrt{3}xz$
	$\mathbb{M}_{2,2}^{(a)}(E_u)$	$\begin{bmatrix} 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 \\ -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{7} \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 \end{bmatrix}$
199	symmetry	$\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{M}_4^{(a)}(A_{1u},1)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{4} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 \\ \frac{i}{4} & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
200	symmetry	$-\frac{\sqrt{15}x^4}{12}+\sqrt{15}x^2y^2-\frac{\sqrt{15}x^2z^2}{2}-\frac{\sqrt{15}y^4}{12}-\frac{\sqrt{15}y^2z^2}{2}+\frac{\sqrt{15}z^4}{6}$
	$\mathbb{M}_4^{(a)}(A_{1u},2)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{210}i}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{210} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{20} & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{84} & 0 \\ -\frac{\sqrt{35}i}{20} & 0 & 0 & 0 & \frac{5\sqrt{21}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
201	symmetry	$\frac{\sqrt{35}x^3y}{2}-\frac{\sqrt{35}xy^3}{2}$
	$\mathbb{M}_4^{(a)}(A_{2u})$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
202	symmetry	$\frac{\sqrt{5}x^4}{4}-\frac{3\sqrt{5}x^2z^2}{2}-\frac{\sqrt{5}y^4}{4}+\frac{3\sqrt{5}y^2z^2}{2}$
	$\mathbb{M}_4^{(a)}(B_{1u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} \\ 0 & \frac{3\sqrt{105}i}{140} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 \\ -\frac{3\sqrt{105}i}{140} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{35} & 0 & 0 & 0 \end{bmatrix}$
203	symmetry	$-\frac{\sqrt{5}x^3y}{2}-\frac{\sqrt{5}xy^3}{2}+3\sqrt{5}xyz^2$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{M}_4^{(a)}(B_{2u})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{105}i}{140} & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & \frac{3\sqrt{105}i}{140} & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{35} & 0 & 0 & 0 & 0 \end{bmatrix}$
204	symmetry	$\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$
	$\mathbb{M}_{4,1}^{(a)}(E_u,1)$	$\begin{bmatrix} \frac{\sqrt{15}i}{40} & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & 0 & \frac{3i}{8} & 0 \\ 0 & 0 & -\frac{3\sqrt{10}i}{40} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ \frac{3\sqrt{5}i}{40} & 0 & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 & 0 \end{bmatrix}$
205	symmetry	$-\frac{\sqrt{35}x^3z}{2} + \frac{\sqrt{35}xz^3}{2}$
	$\mathbb{M}_{4,2}^{(a)}(E_u,1)$	$\begin{bmatrix} 0 & \frac{\sqrt{15}i}{40} & 0 & 0 & 0 & -\frac{i}{8} & 0 \\ -\frac{\sqrt{15}i}{40} & 0 & 0 & 0 & \frac{3i}{8} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{10}i}{40} & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} \\ 0 & -\frac{3\sqrt{5}i}{40} & 0 & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 \end{bmatrix}$
206	symmetry	$3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$
	$\mathbb{M}_{4,1}^{(a)}(E_u,2)$	$\begin{bmatrix} \frac{\sqrt{105}i}{280} & 0 & 0 & 0 & \frac{9\sqrt{7}i}{56} & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{280} & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{40} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{35}i}{40} & 0 & 0 & 0 & \frac{\sqrt{21}i}{56} & 0 & 0 \end{bmatrix}$
207	symmetry	$\frac{\sqrt{5}x^3z}{2} - 3\sqrt{5}xy^2z + \frac{\sqrt{5}xz^3}{2}$

continued ...

Table 9

No.	multipole	matrix
	$\mathbb{M}^{(a)}_{4,2}(E_u,2)$	$\begin{bmatrix} 0 & \frac{\sqrt{105}i}{280} & 0 & 0 & 0 & -\frac{9\sqrt{7}i}{56} & 0 \\ -\frac{\sqrt{105}i}{280} & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{40} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} \\ 0 & \frac{3\sqrt{35}i}{40} & 0 & 0 & 0 & \frac{\sqrt{21}i}{56} & 0 \end{bmatrix}$

bra: $= \langle f_c| \, , \langle f_s| \, , \langle f_z^\beta| \, , \langle f_{xyz}| \, , \langle f_x| \, , \langle f_y| \, , \langle f_z^\alpha|$
ket: $= |f_c\rangle \, , |f_s\rangle \, , |f_z^\beta\rangle \, , |f_{xyz}\rangle \, , |f_x\rangle \, , |f_y\rangle \, , |f_z^\alpha\rangle$

Table 10: (f,f) block.

No.	multipole	matrix
208	symmetry	1
	$\mathbb{Q}^{(a)}_0(A_{1g})$	$\begin{bmatrix} \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} \end{bmatrix}$
209	symmetry	$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$
	$\mathbb{Q}^{(a)}_2(A_{1g})$	$\begin{bmatrix} -\frac{5\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{21} \end{bmatrix}$
210	symmetry	$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_2^{(a)}(B_{1g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{21} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{21} & 0 & 0 & 0 & 0 \end{bmatrix}$
211	symmetry	$\sqrt{3}xy$
	$\mathbb{Q}_2^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{21} \\ 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 \\ \frac{\sqrt{105}}{42} & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{21} & 0 & 0 & 0 \end{bmatrix}$
212	symmetry	$\sqrt{3}yz$
	$\mathbb{Q}_{2,1}^{(a)}(E_g)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{5\sqrt{42}}{84} & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{42}}{84} & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{42}}{84} & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 \\ -\frac{5\sqrt{42}}{84} & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & \frac{\sqrt{42}}{42} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 \end{bmatrix}$
213	symmetry	$\sqrt{3}xz$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}^{(a)}_{2,2}(E_g)$	$\begin{bmatrix} 0 & 0 & \frac{5\sqrt{42}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{42}}{84} & 0 & 0 & 0 \\ \frac{5\sqrt{42}}{84} & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 \\ 0 & \frac{5\sqrt{42}}{84} & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 \\ 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & \frac{\sqrt{42}}{42} \\ 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 \end{bmatrix}$
214	symmetry	$\frac{\sqrt{21}x^4}{6} - \frac{\sqrt{21}x^2y^2}{2} - \frac{\sqrt{21}x^2z^2}{2} + \frac{\sqrt{21}y^4}{6} - \frac{\sqrt{21}y^2z^2}{2} + \frac{\sqrt{21}z^4}{6}$
	$\mathbb{Q}^{(a)}_{4}(A_{1g},1)$	$\begin{bmatrix} \frac{\sqrt{66}}{44} & 0 & 0 & 0 & -\frac{\sqrt{110}}{44} & 0 & 0 \\ 0 & \frac{\sqrt{66}}{44} & 0 & 0 & 0 & \frac{\sqrt{110}}{44} & 0 \\ 0 & 0 & -\frac{\sqrt{66}}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{66}}{11} & 0 & 0 & 0 \\ -\frac{\sqrt{110}}{44} & 0 & 0 & 0 & \frac{\sqrt{66}}{132} & 0 & 0 \\ 0 & \frac{\sqrt{110}}{44} & 0 & 0 & 0 & \frac{\sqrt{66}}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{22} \end{bmatrix}$
215	symmetry	$-\frac{\sqrt{15}x^4}{12} + \sqrt{15}x^2y^2 - \frac{\sqrt{15}x^2z^2}{2} - \frac{\sqrt{15}y^4}{12} - \frac{\sqrt{15}y^2z^2}{2} + \frac{\sqrt{15}z^4}{6}$
	$\mathbb{Q}^{(a)}_{4}(A_{1g},2)$	$\begin{bmatrix} \frac{\sqrt{2310}}{308} & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 \\ 0 & \frac{\sqrt{2310}}{308} & 0 & 0 & 0 & -\frac{\sqrt{154}}{44} & 0 \\ 0 & 0 & -\frac{\sqrt{2310}}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{154}}{44} & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} & 0 & 0 \\ 0 & -\frac{\sqrt{154}}{44} & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{154} \end{bmatrix}$
216	symmetry	$\frac{\sqrt{35}x^3y}{2} - \frac{\sqrt{35}xy^3}{2}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_4^{(a)}(A_{2g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{110}}{22} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{110}}{22} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
217	symmetry	$\frac{\sqrt{5}x^4}{4} - \frac{3\sqrt{5}x^2z^2}{2} - \frac{\sqrt{5}y^4}{4} + \frac{3\sqrt{5}y^2z^2}{2}$
	$\mathbb{Q}_4^{(a)}(B_{1g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{462}}{154} & 0 & 0 & 0 & -\frac{\sqrt{770}}{77} & 0 & 0 \\ 0 & -\frac{3\sqrt{462}}{154} & 0 & 0 & 0 & \frac{\sqrt{770}}{77} & 0 \\ 0 & 0 & \frac{\sqrt{462}}{154} & 0 & 0 & 0 & 0 \end{bmatrix}$
218	symmetry	$-\frac{\sqrt{5}x^3y}{2} - \frac{\sqrt{5}xy^3}{2} + 3\sqrt{5}xyz^2$
	$\mathbb{Q}_4^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{154} & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{462}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{154} \\ 0 & \frac{3\sqrt{462}}{154} & 0 & 0 & 0 & \frac{\sqrt{770}}{77} & 0 \\ -\frac{3\sqrt{462}}{154} & 0 & 0 & 0 & \frac{\sqrt{770}}{77} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{462}}{154} & 0 & 0 & 0 \end{bmatrix}$
219	symmetry	$\frac{\sqrt{35}y^3z}{2} - \frac{\sqrt{35}yz^3}{2}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_{4,1}^{(a)}(E_g,1)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{165}}{44} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{165}}{44} & 0 & 0 & 0 & \frac{3\sqrt{11}}{44} \\ 0 & \frac{\sqrt{165}}{44} & 0 & 0 & 0 & \frac{3\sqrt{11}}{44} & 0 \\ -\frac{\sqrt{165}}{44} & 0 & 0 & 0 & -\frac{5\sqrt{11}}{44} & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{11}}{44} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{11}}{44} & 0 & 0 & 0 & -\frac{\sqrt{165}}{44} \\ 0 & \frac{3\sqrt{11}}{44} & 0 & 0 & 0 & -\frac{\sqrt{165}}{44} & 0 \end{bmatrix}$
220	symmetry	$\frac{\sqrt{35}x^3z}{2} - \frac{\sqrt{35}xz^3}{2}$
	$\mathbb{Q}_{4,2}^{(a)}(E_g,1)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{165}}{44} & 0 & 0 & 0 & -\frac{3\sqrt{11}}{44} \\ 0 & 0 & 0 & \frac{\sqrt{165}}{44} & 0 & 0 & 0 \\ \frac{\sqrt{165}}{44} & 0 & 0 & 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 \\ 0 & \frac{\sqrt{165}}{44} & 0 & 0 & 0 & -\frac{5\sqrt{11}}{44} & 0 \\ 0 & 0 & -\frac{3\sqrt{11}}{44} & 0 & 0 & 0 & -\frac{\sqrt{165}}{44} \\ 0 & 0 & 0 & -\frac{5\sqrt{11}}{44} & 0 & 0 & 0 \\ -\frac{3\sqrt{11}}{44} & 0 & 0 & 0 & -\frac{\sqrt{165}}{44} & 0 & 0 \end{bmatrix}$
221	symmetry	$3\sqrt{5}x^2yz - \frac{\sqrt{5}y^3z}{2} - \frac{\sqrt{5}yz^3}{2}$
	$\mathbb{Q}_{4,1}^{(a)}(E_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{1155}}{308} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & -\frac{3\sqrt{77}}{44} \\ 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & \frac{\sqrt{77}}{28} & 0 \\ -\frac{\sqrt{1155}}{308} & 0 & 0 & 0 & \frac{3\sqrt{77}}{308} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{77}}{308} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{77}}{28} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{308} \\ 0 & -\frac{3\sqrt{77}}{44} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{308} & 0 \end{bmatrix}$
222	symmetry	$-\frac{\sqrt{5}x^3z}{2} + 3\sqrt{5}xy^2z - \frac{\sqrt{5}xz^3}{2}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_{4,2}^{(a)}(E_g,2)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & \frac{3\sqrt{77}}{44} \\ 0 & 0 & 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 \\ \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & -\frac{\sqrt{77}}{28} & 0 & 0 \\ 0 & \frac{\sqrt{1155}}{308} & 0 & 0 & 0 & \frac{3\sqrt{77}}{308} & 0 \\ 0 & 0 & -\frac{\sqrt{77}}{28} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{308} \\ 0 & 0 & 0 & \frac{3\sqrt{77}}{308} & 0 & 0 & 0 \\ \frac{3\sqrt{77}}{44} & 0 & 0 & 0 & -\frac{\sqrt{1155}}{308} & 0 & 0 \end{bmatrix}$
223	symmetry	$\frac{\sqrt{2}x^6}{4} - \frac{15\sqrt{2}x^4y^2}{8} - \frac{15\sqrt{2}x^4z^2}{8} - \frac{15\sqrt{2}x^2y^4}{8} + \frac{45\sqrt{2}x^2y^2z^2}{2} - \frac{15\sqrt{2}x^2z^4}{8} + \frac{\sqrt{2}y^6}{4} - \frac{15\sqrt{2}y^4z^2}{8} - \frac{15\sqrt{2}y^2z^4}{8} + \frac{\sqrt{2}z^6}{4}$
	$\mathbb{Q}_6^{(a)}(A_{1g},1)$	$\begin{bmatrix} -\frac{\sqrt{462}}{1848} & 0 & 0 & 0 & -\frac{\sqrt{770}}{88} & 0 & 0 \\ 0 & -\frac{\sqrt{462}}{1848} & 0 & 0 & 0 & \frac{\sqrt{770}}{88} & 0 \\ 0 & 0 & -\frac{3\sqrt{462}}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{462}}{77} & 0 & 0 & 0 \\ -\frac{\sqrt{770}}{88} & 0 & 0 & 0 & -\frac{5\sqrt{462}}{616} & 0 & 0 \\ 0 & \frac{\sqrt{770}}{88} & 0 & 0 & 0 & -\frac{5\sqrt{462}}{616} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{462} \end{bmatrix}$
224	symmetry	$-\frac{\sqrt{14}x^6}{8} + \frac{15\sqrt{14}x^4z^2}{8} - \frac{15\sqrt{14}x^2z^4}{8} - \frac{\sqrt{14}y^6}{8} + \frac{15\sqrt{14}y^4z^2}{8} - \frac{15\sqrt{14}y^2z^4}{8} + \frac{\sqrt{14}z^6}{4}$
	$\mathbb{Q}_6^{(a)}(A_{1g},2)$	$\begin{bmatrix} -\frac{\sqrt{66}}{264} & 0 & 0 & 0 & \frac{\sqrt{110}}{88} & 0 & 0 \\ 0 & -\frac{\sqrt{66}}{264} & 0 & 0 & 0 & -\frac{\sqrt{110}}{88} & 0 \\ 0 & 0 & \frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{110}}{88} & 0 & 0 & 0 & -\frac{5\sqrt{66}}{88} & 0 & 0 \\ 0 & -\frac{\sqrt{110}}{88} & 0 & 0 & 0 & -\frac{5\sqrt{66}}{88} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{66} \end{bmatrix}$
225	symmetry	$-\frac{3\sqrt{7}x^5y}{4} + \frac{15\sqrt{7}x^3yz^2}{2} + \frac{3\sqrt{7}xy^5}{4} - \frac{15\sqrt{7}xy^3z^2}{2}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_6^{(a)}(A_{2g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{33}}{11} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{33}}{11} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
226	symmetry	$-\frac{\sqrt{2310}x^4y^2}{8} + \frac{\sqrt{2310}x^4z^2}{8} + \frac{\sqrt{2310}x^2y^4}{8} - \frac{\sqrt{2310}x^2z^4}{8} - \frac{\sqrt{2310}y^4z^2}{8} + \frac{\sqrt{2310}y^2z^4}{8}$
	$\mathbb{Q}_6^{(a)}(B_{1g},1)$	$\begin{bmatrix} \frac{\sqrt{10}}{8} & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{8} & 0 & 0 & 0 & \frac{\sqrt{6}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{24} & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{24} & 0 & 0 & 0 & \frac{\sqrt{10}}{8} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \end{bmatrix}$
227	symmetry	$\frac{\sqrt{42}x^6}{8} - \frac{5\sqrt{42}x^4y^2}{4} - \frac{5\sqrt{42}x^4z^2}{8} + \frac{5\sqrt{42}x^2y^4}{4} + \frac{5\sqrt{42}x^2z^4}{8} - \frac{\sqrt{42}y^6}{8} + \frac{5\sqrt{42}y^4z^2}{8} - \frac{5\sqrt{42}y^2z^4}{8}$
	$\mathbb{Q}_6^{(a)}(B_{1g},2)$	$\begin{bmatrix} \frac{\sqrt{22}}{8} & 0 & 0 & 0 & -\frac{\sqrt{330}}{264} & 0 & 0 \\ 0 & -\frac{\sqrt{22}}{8} & 0 & 0 & 0 & -\frac{\sqrt{330}}{264} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{330}}{264} & 0 & 0 & 0 & \frac{5\sqrt{22}}{88} & 0 & 0 \\ 0 & -\frac{\sqrt{330}}{264} & 0 & 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 \\ 0 & 0 & \frac{\sqrt{330}}{66} & 0 & 0 & 0 & 0 \end{bmatrix}$
228	symmetry	$\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_6^{(a)}(B_{2g},1)$	$\begin{bmatrix} 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
229	symmetry	$\frac{\sqrt{210}x^5y}{16}+\frac{\sqrt{210}x^3y^3}{8}-\sqrt{210}x^3yz^2+\frac{\sqrt{210}xy^5}{16}-\sqrt{210}xy^3z^2+\sqrt{210}xyz^4$
	$\mathbb{Q}_6^{(a)}(B_{2g},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{66} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{66} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{66}}{33} \\ 0 & -\frac{\sqrt{66}}{66} & 0 & 0 & 0 & \frac{\sqrt{110}}{22} & 0 \\ \frac{\sqrt{66}}{66} & 0 & 0 & 0 & \frac{\sqrt{110}}{22} & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{66}}{33} & 0 & 0 & 0 \end{bmatrix}$
230	symmetry	$\frac{15\sqrt{7}x^2y^3z}{2}-\frac{15\sqrt{7}x^2yz^3}{2}-\frac{3\sqrt{7}y^5z}{4}+\frac{3\sqrt{7}yz^5}{4}$
	$\mathbb{Q}_{6,1}^{(a)}(E_g,1)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{3\sqrt{22}}{44} & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} \\ 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & -\frac{\sqrt{330}}{88} & 0 \\ -\frac{3\sqrt{22}}{44} & 0 & 0 & 0 & -\frac{\sqrt{330}}{44} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{330}}{44} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & \frac{5\sqrt{22}}{88} \\ 0 & -\frac{\sqrt{330}}{88} & 0 & 0 & 0 & \frac{5\sqrt{22}}{88} & 0 \end{bmatrix}$
231	symmetry	$-\frac{3\sqrt{7}x^5z}{4}+\frac{15\sqrt{7}x^3y^2z}{2}-\frac{15\sqrt{7}xy^2z^3}{2}+\frac{3\sqrt{7}xz^5}{4}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_{6,2}^{(a)}(E_g,1)$	$\begin{bmatrix} 0 & 0 & -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & \frac{\sqrt{330}}{88} \\ 0 & 0 & 0 & \frac{3\sqrt{22}}{44} & 0 & 0 & 0 \\ -\frac{5\sqrt{22}}{88} & 0 & 0 & 0 & \frac{\sqrt{330}}{88} & 0 & 0 \\ 0 & \frac{3\sqrt{22}}{44} & 0 & 0 & 0 & -\frac{\sqrt{330}}{44} & 0 \\ 0 & 0 & \frac{\sqrt{330}}{88} & 0 & 0 & 0 & \frac{5\sqrt{22}}{88} \\ 0 & 0 & 0 & -\frac{\sqrt{330}}{44} & 0 & 0 & 0 \\ \frac{\sqrt{330}}{88} & 0 & 0 & 0 & \frac{5\sqrt{22}}{88} & 0 & 0 \end{bmatrix}$
232	symmetry	$\frac{3\sqrt{462}y^5z}{16} - \frac{5\sqrt{462}y^3z^3}{8} + \frac{3\sqrt{462}yz^5}{16}$
	$\mathbb{Q}_{6,1}^{(a)}(E_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{16} & 0 & 0 & 0 & \frac{\sqrt{5}}{16} \\ 0 & \frac{\sqrt{3}}{16} & 0 & 0 & 0 & \frac{3\sqrt{5}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{5}}{16} & 0 & 0 & 0 & \frac{5\sqrt{3}}{16} \\ 0 & \frac{\sqrt{5}}{16} & 0 & 0 & 0 & \frac{5\sqrt{3}}{16} & 0 \end{bmatrix}$
233	symmetry	$\frac{3\sqrt{462}x^5z}{16} - \frac{5\sqrt{462}x^3z^3}{8} + \frac{3\sqrt{462}xz^5}{16}$
	$\mathbb{Q}_{6,2}^{(a)}(E_g,2)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{16} & 0 & 0 & 0 & -\frac{\sqrt{5}}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}}{16} & 0 & 0 & 0 & -\frac{3\sqrt{5}}{16} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{5}}{16} & 0 & 0 & 0 & \frac{5\sqrt{3}}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{16} & 0 & 0 & 0 & \frac{5\sqrt{3}}{16} & 0 & 0 \end{bmatrix}$
234	symmetry	$\sqrt{210}x^4yz - \sqrt{210}x^2y^3z - \sqrt{210}x^2yz^3 + \frac{\sqrt{210}y^5z}{16} + \frac{\sqrt{210}y^3z^3}{8} + \frac{\sqrt{210}yz^5}{16}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{Q}_{6,1}^{(a)}(E_g,3)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{165}}{33} & 0 & 0 & 0 \\ 0 & 0 & \frac{17\sqrt{165}}{528} & 0 & 0 & 0 & -\frac{9\sqrt{11}}{176} \\ 0 & \frac{17\sqrt{165}}{528} & 0 & 0 & 0 & -\frac{\sqrt{11}}{16} & 0 \\ \frac{\sqrt{165}}{33} & 0 & 0 & 0 & -\frac{\sqrt{11}}{11} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{11}}{11} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{11}}{16} & 0 & 0 & 0 & \frac{5\sqrt{165}}{528} \\ 0 & -\frac{9\sqrt{11}}{176} & 0 & 0 & 0 & \frac{5\sqrt{165}}{528} & 0 \end{bmatrix}$
235	symmetry	$\frac{\sqrt{210}x^5z}{16} - \sqrt{210}x^3y^2z + \frac{\sqrt{210}x^3z^3}{8} + \sqrt{210}xy^4z - \sqrt{210}xy^2z^3 + \frac{\sqrt{210}xz^5}{16}$
	$\mathbb{Q}_{6,2}^{(a)}(E_g,3)$	$\begin{bmatrix} 0 & 0 & \frac{17\sqrt{165}}{528} & 0 & 0 & 0 & \frac{9\sqrt{11}}{176} \\ 0 & 0 & 0 & -\frac{\sqrt{165}}{33} & 0 & 0 & 0 \\ \frac{17\sqrt{165}}{528} & 0 & 0 & 0 & \frac{\sqrt{11}}{16} & 0 & 0 \\ 0 & -\frac{\sqrt{165}}{33} & 0 & 0 & 0 & -\frac{\sqrt{11}}{11} & 0 \\ 0 & 0 & \frac{\sqrt{11}}{16} & 0 & 0 & 0 & \frac{5\sqrt{165}}{528} \\ 0 & 0 & 0 & -\frac{\sqrt{11}}{11} & 0 & 0 & 0 \\ \frac{9\sqrt{11}}{176} & 0 & 0 & 0 & \frac{5\sqrt{165}}{528} & 0 & 0 \end{bmatrix}$
236	symmetry	z
	$\mathbb{M}_1^{(a)}(A_{2g})$	$\begin{bmatrix} 0 & -\frac{3\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{7} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
237	symmetry	x

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}^{(a)}_{1,1}(E_g)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 \\ -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 \end{bmatrix}$
238	symmetry	$-y$
	$\mathbb{M}^{(a)}_{1,2}(E_g)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 \end{bmatrix}$
239	symmetry	$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$
	$\mathbb{M}^{(a)}_{3}(A_{2g})$	$\begin{bmatrix} 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
240	symmetry	$\sqrt{15}xyz$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}_{3}^{(a)}(B_{1g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \end{bmatrix}$
241	symmetry	$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$
	$\mathbb{M}_{3}^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \end{bmatrix}$
242	symmetry	$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$
	$\mathbb{M}_{3,1}^{(a)}(E_g, 1)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} \\ 0 & \frac{i}{4} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 \\ -\frac{i}{4} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & \frac{i}{4} \\ 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & -\frac{i}{4} & 0 \end{bmatrix}$
243	symmetry	$\frac{3x^2y}{2} - y^3 + \frac{3yz^2}{2}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}^{(a)}_{3,2}(E_g,1)$	$\begin{bmatrix} 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} \\ 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 \\ \frac{i}{4} & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 \\ 0 & \frac{i}{4} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & \frac{i}{4} \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 \end{bmatrix}$
244	symmetry	$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$
	$\mathbb{M}^{(a)}_{3,1}(E_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & -\frac{i}{4} \\ 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & \frac{i}{4} & 0 \\ -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} \\ 0 & \frac{i}{4} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 \end{bmatrix}$
245	symmetry	$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$
	$\mathbb{M}^{(a)}_{3,2}(E_g,2)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & \frac{i}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & \frac{i}{4} & 0 \\ 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} \\ 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 \\ -\frac{i}{4} & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 \end{bmatrix}$
246	symmetry	$\frac{3\sqrt{35}x^3yz}{2} - \frac{3\sqrt{35}xy^3z}{2}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}_5^{(a)}(A_{1g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
247	symmetry	$\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$
	$\mathbb{M}_5^{(a)}(A_{2g},1)$	$\begin{bmatrix} 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}i}{21} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{42} & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
248	symmetry	$\frac{3\sqrt{35}x^4z}{8}-\frac{9\sqrt{35}x^2y^2z}{4}+\frac{3\sqrt{35}y^4z}{8}$
	$\mathbb{M}_5^{(a)}(A_{2g},2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
249	symmetry	$\frac{\sqrt{105}x^3yz}{2}+\frac{\sqrt{105}xy^3z}{2}-\sqrt{105}xyz^3$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}_5^{(a)}(B_{1g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{3} & 0 & 0 & 0 & 0 \end{bmatrix}$
250	symmetry	$-\frac{\sqrt{105}x^4z}{4} + \frac{\sqrt{105}x^2z^3}{2} + \frac{\sqrt{105}y^4z}{4} - \frac{\sqrt{105}y^2z^3}{2}$
	$\mathbb{M}_5^{(a)}(B_{2g})$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{3} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{3} & 0 & 0 & 0 \end{bmatrix}$
251	symmetry	$x^5 - 5x^3y^2 - 5x^3z^2 + \frac{15xy^4}{8} + \frac{15xy^2z^2}{4} + \frac{15xz^4}{8}$
	$\mathbb{M}_{5,1}^{(a)}(E_g, 1)$	$\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{13\sqrt{14}i}{112} & 0 & 0 & 0 & \frac{\sqrt{210}i}{48} \\ 0 & \frac{13\sqrt{14}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{336} & 0 \\ \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & \frac{\sqrt{210}i}{42} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{112} \\ 0 & -\frac{\sqrt{210}i}{48} & 0 & 0 & 0 & \frac{5\sqrt{14}i}{112} & 0 \end{bmatrix}$
252	symmetry	$-\frac{15x^4y}{8} + 5x^2y^3 - \frac{15x^2yz^2}{4} - y^5 + 5y^3z^2 - \frac{15yz^4}{8}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}^{(a)}_{5,2}(E_g,1)$	$\begin{bmatrix} 0 & 0 & -\frac{13\sqrt{14}i}{112} & 0 & 0 & 0 & -\frac{\sqrt{210}i}{48} \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ \frac{13\sqrt{14}i}{112} & 0 & 0 & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & \frac{\sqrt{210}i}{42} & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{336} & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{112} \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 \\ \frac{\sqrt{210}i}{48} & 0 & 0 & 0 & \frac{5\sqrt{14}i}{112} & 0 & 0 \end{bmatrix}$
253	symmetry	$\frac{3\sqrt{35}xy^4}{8} - \frac{9\sqrt{35}xy^2z^2}{4} + \frac{3\sqrt{35}xz^4}{8}$
	$\mathbb{M}^{(a)}_{5,1}(E_g,2)$	$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{16} \\ 0 & \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{6}i}{16} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} \\ 0 & \frac{3\sqrt{6}i}{16} & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 \end{bmatrix}$
254	symmetry	$-\frac{3\sqrt{35}x^4y}{8} + \frac{9\sqrt{35}x^2yz^2}{4} - \frac{3\sqrt{35}yz^4}{8}$
	$\mathbb{M}^{(a)}_{5,2}(E_g,2)$	$\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}i}{16} & 0 & 0 & 0 & \frac{3\sqrt{6}i}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{16} & 0 & 0 & 0 & \frac{3\sqrt{6}i}{16} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{6}i}{16} & 0 & 0 & 0 & -\frac{\sqrt{10}i}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{6}i}{16} & 0 & 0 & 0 & \frac{\sqrt{10}i}{16} & 0 & 0 \end{bmatrix}$
255	symmetry	$\frac{\sqrt{105}x^3y^2}{2} - \frac{\sqrt{105}x^3z^2}{2} - \frac{\sqrt{105}xy^4}{4} + \frac{\sqrt{105}xz^4}{4}$

continued ...

Table 10

No.	multipole	matrix
	$\mathbb{M}^{(a)}_{5,1}(E_g,3)$	$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{30}i}{12} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} \\ 0 & -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 \\ -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} \\ 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 \end{bmatrix}$
256	symmetry	$\frac{\sqrt{105}x^4y}{4}-\frac{\sqrt{105}x^2y^3}{2}+\frac{\sqrt{105}y^3z^2}{2}-\frac{\sqrt{105}yz^4}{4}$
	$\mathbb{M}^{(a)}_{5,2}(E_g,3)$	$\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{24} & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{12} & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 \end{bmatrix}$